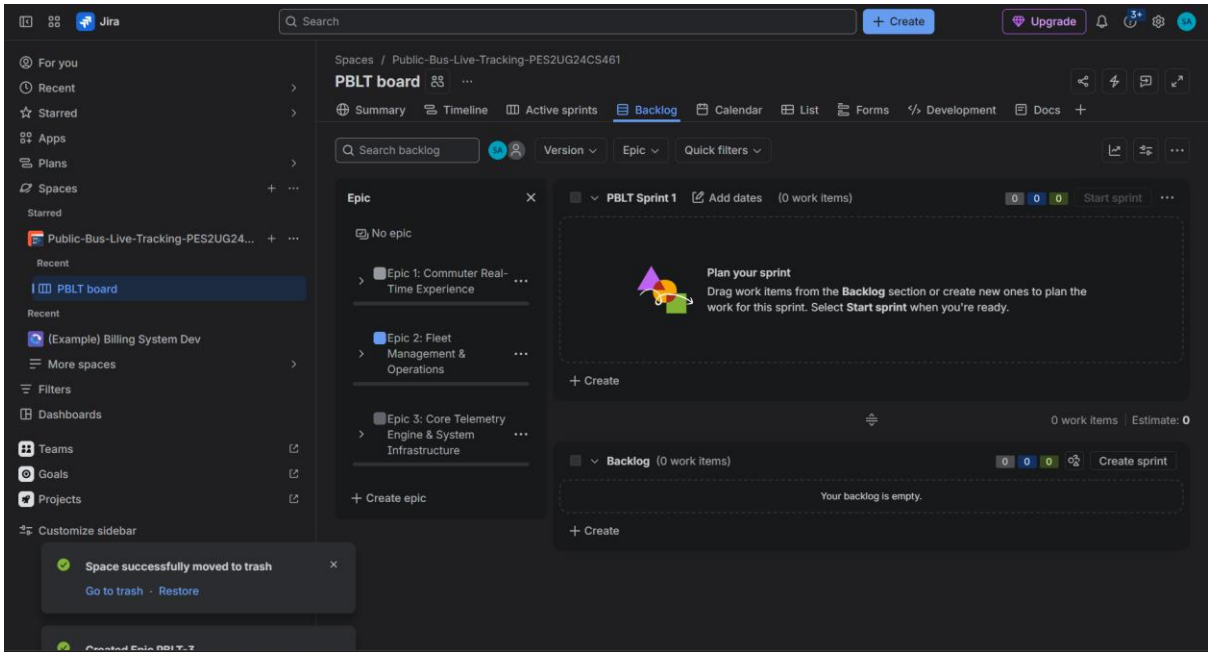
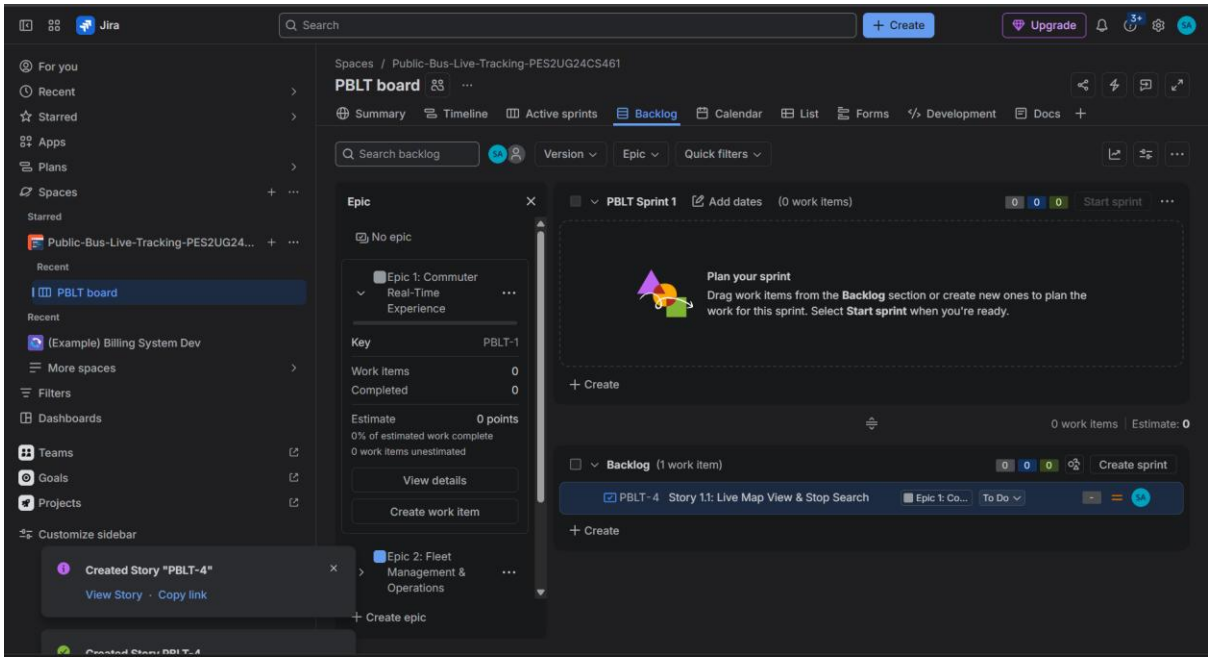
Software Engineering Lab-2

PES2UG24CS461 – Public Bus Live Tracking and Crowding Estimator

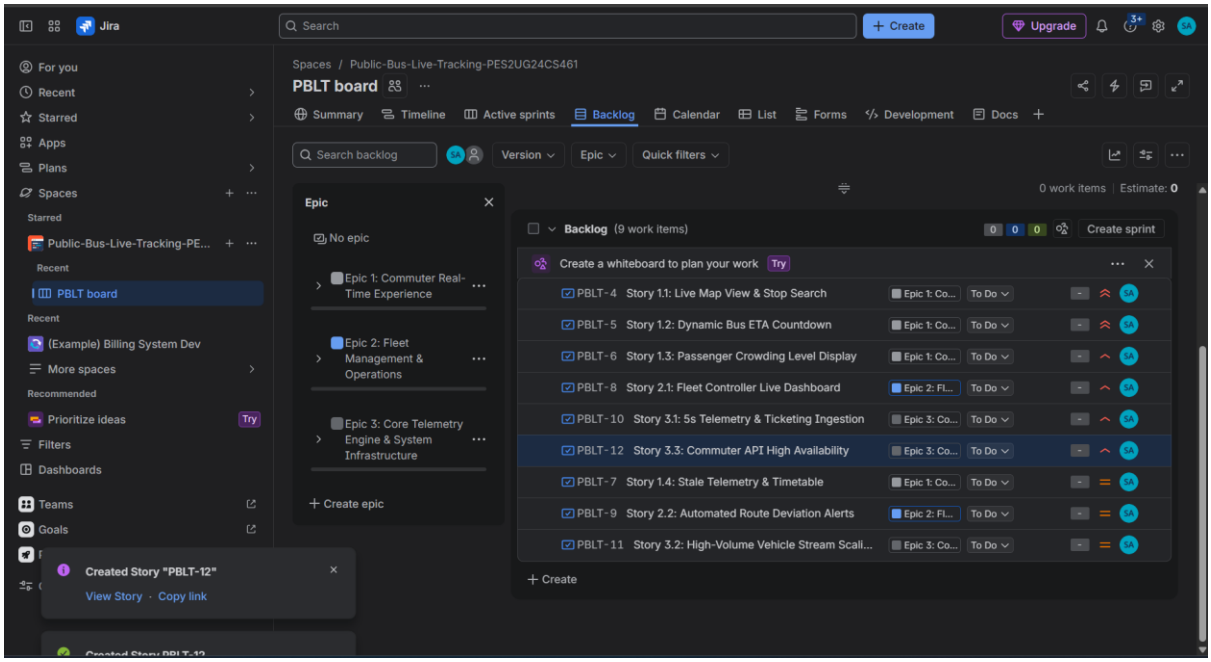
1) Created Public-Bus-Live-Tracking Space. Then added 3 epics:
    a. Commuter Real-Time Experience
    b. Fleet Management and operations
    c. Core Telemetry and System Infrastructure



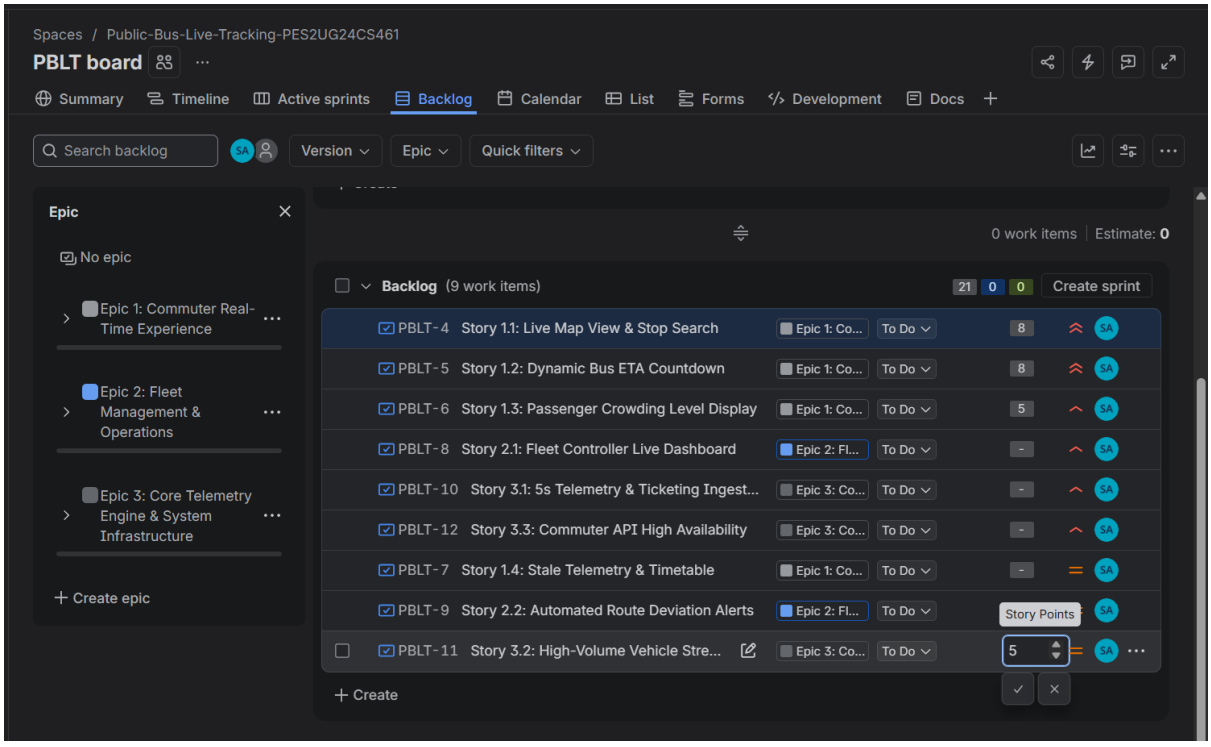
2) Addition of User Story for the first Epic after creating 3 Epics



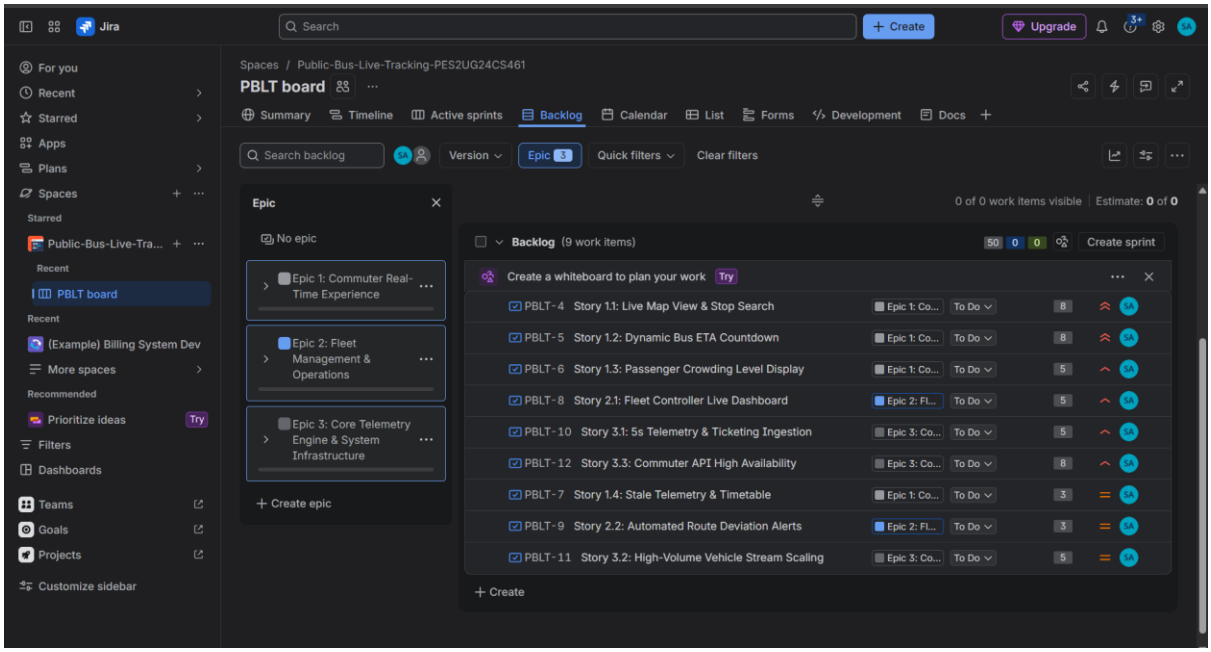
3) Added User Stories for each of the 3 Epics (Total of 9 User Stories)



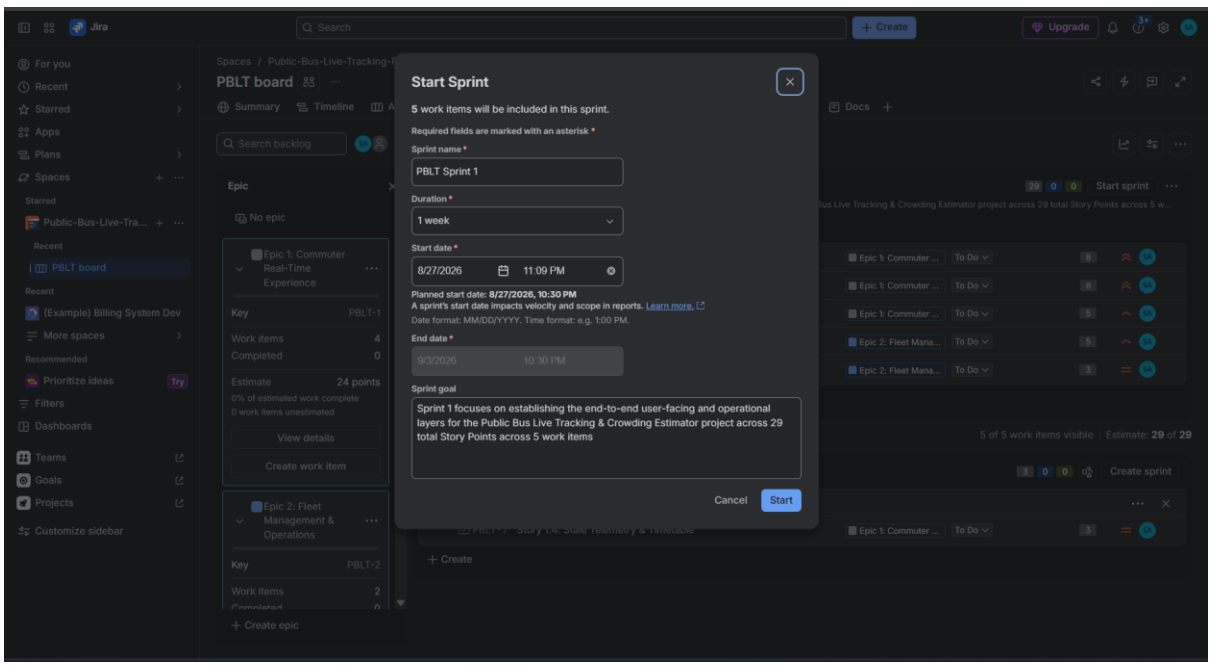
4) Adding Story Points to Each of the User Stories based on the Effort taken to be fully implemented and its importance



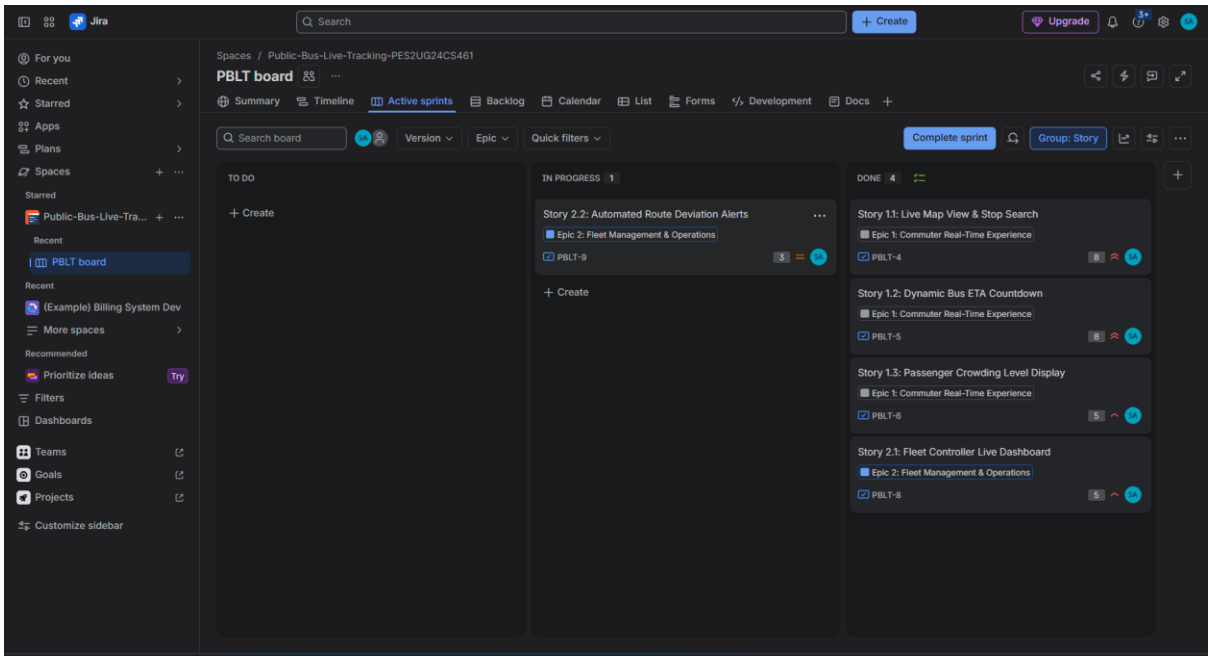
5) Complete Board with a total of 3 Epics with 9 User Stories arranged with Story Points and Priorities – Hardest, Hard and Medium



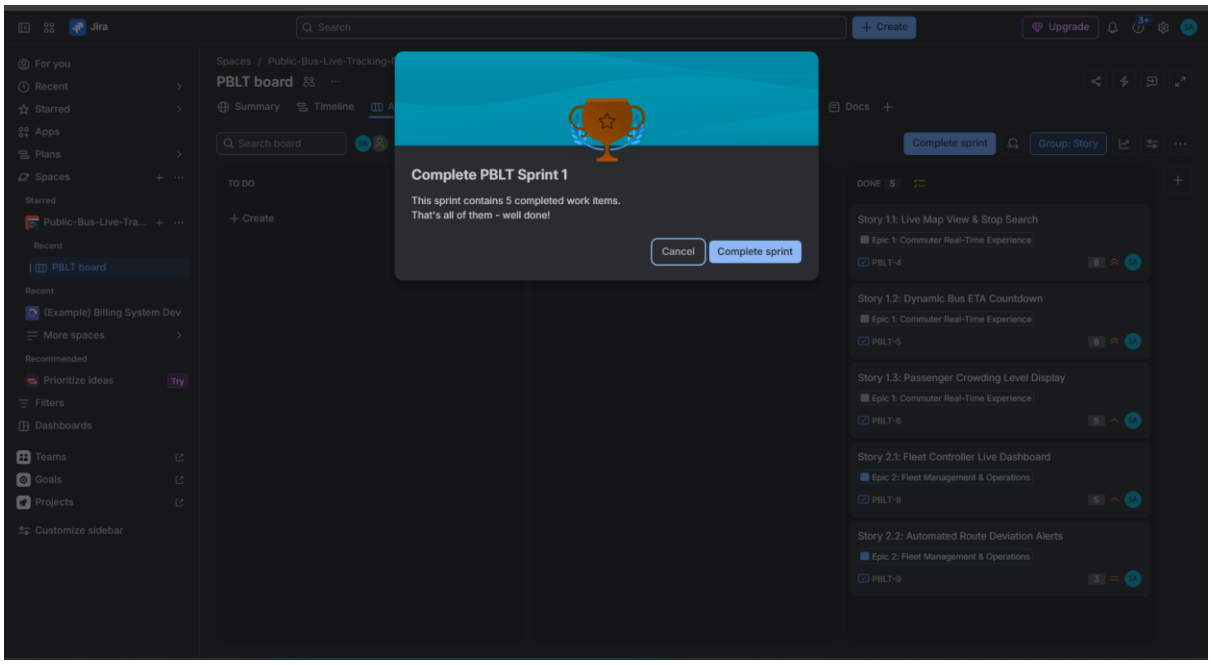
6) Creation of Sprint 1 with selective User Stories that require attention first (Core, Critical Backbone Features required before the Analytics and Enhancements) before the other user stories of lesser importance are implemented



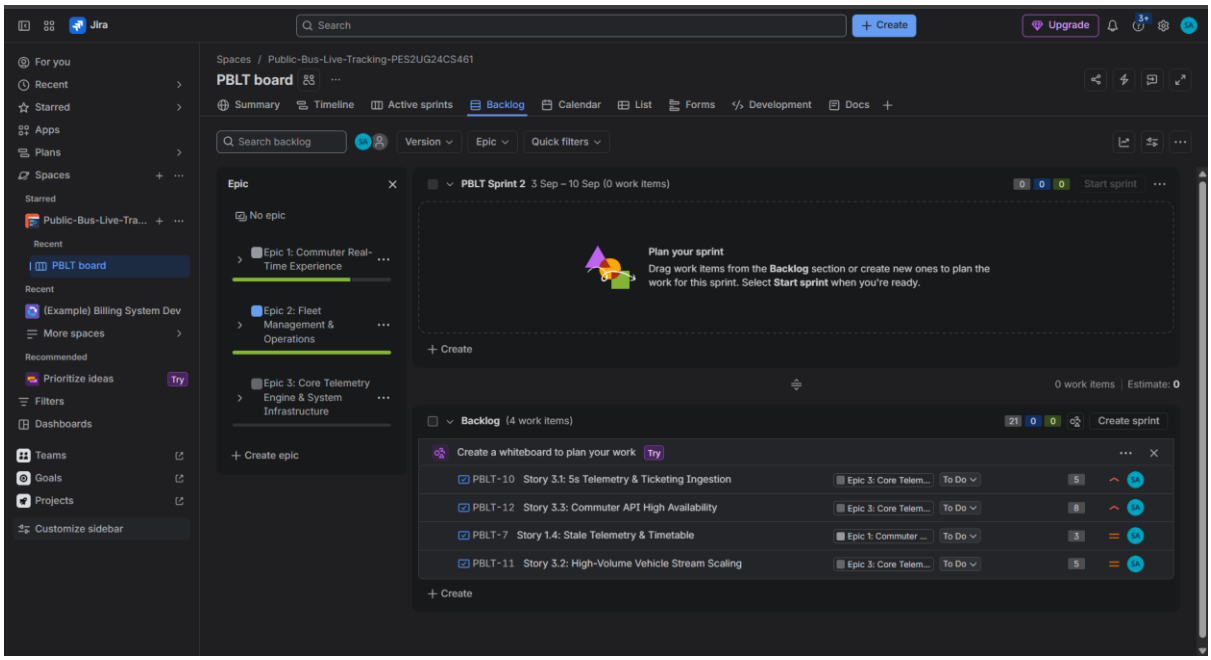
7) Completion of User Stories over time based on Importance/Priority and Available resources and Story Points



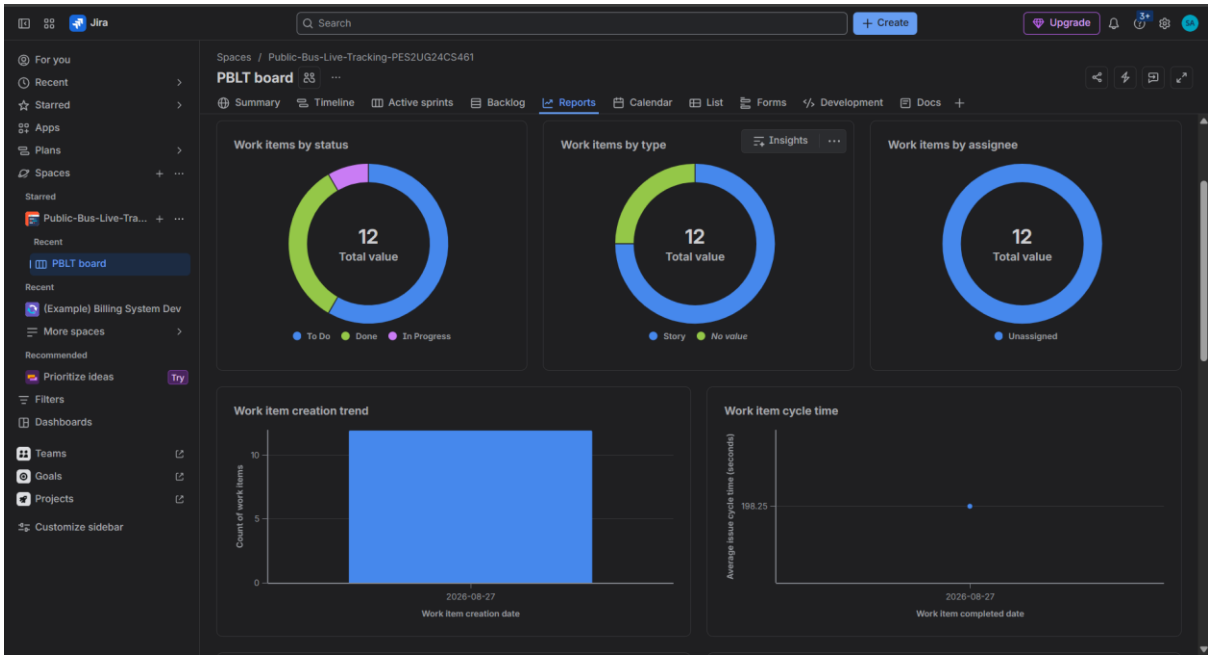
8) Completed all the User Story Tasks and marked the Sprint 1 as Complete



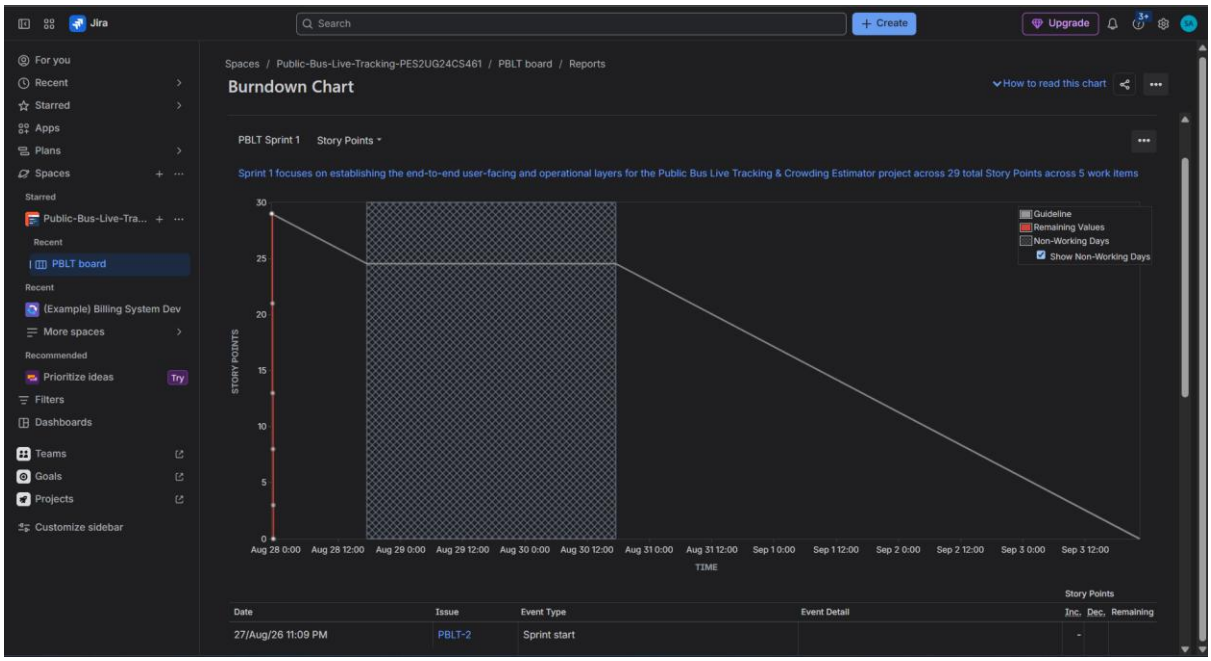
9) Completed Sprint 1



10) Report Metrics



11) Burndown Chart for Sprint 1 showing Each User Story – From Todo -> In-Progress -> Done (Story Points v/s Time Plot1)



Reflection Questions:

1. Did your estimations reflect the actual effort?

   Yes, using the Fibonacci sequence (3, 5, 8) accurately captured the relative complexity of each feature. Higher effort was assigned to complex core logic like the live map tracking and dynamic ETA countdown of 8 points each, while simpler UI tasks like passenger crowding badges and route deviation alerts were appropriately estimated at 5 and 3 points.

2. Was your backlog well-prioritized?

   Yes, the backlog was prioritized by business value and dependencies. The foundational commuter-facing features and fleet monitoring dashboard (PBLT-4, 5, 6, 8, 9) were given high priority and scheduled into Sprint 1 (29 points), while infrastructure scaling, API availability, and timetable fallbacks were deferred to Sprint 2 (21 points).

3. How did your simulated sprint align with your plan?

   The simulated sprint aligned well with the goal of establishing the end-to-end live tracking system. All 5 planned user stories progressed from To Do through In Progress to Done, achieving 100% completion of the committed 29 story points without leaving any scope unfinished.

4. What insights did the burndown chart give about your team's capacity?

   The chart shows a sprint capacity of 29 story points across 5 work items. Because the tasks were updated within a single lab simulation session, the remaining work line drops sharply down to zero on Day 1 rather than steadily tracking along the grey guideline across the full 1-week timeline. In a real sprint, this pattern highlights the need to close tasks incrementally day-by-day to maintain an even workflow.